

Markscheme

Mathematics: analysis and approaches

Practice question bank

156. (a)

A cubic equation with real coefficients has roots 2 , $1 + 2i$, and $1 - 2i$.

Find the cubic equation in the form $x^3 + px^2 + qx + r = 0$, stating the values of p , q and r .

multiply the complex conjugate pair first: $(x - (1 + 2i))(x - (1 - 2i)) = (x - 1)^2 + 4 = x^2 - 2x + 5$ (M1) A1

multiply by $(x - 2)$: $(x - 2)(x^2 - 2x + 5)$ (M1)

$$= x^3 - 4x^2 + 9x - 10 \quad \text{A1}$$

$$p = -4, \quad q = 9, \quad r = -10 \quad \text{A1}$$

[6 marks]